Robin Belton

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Academic Appointments

Vassar College Poughkeepsie, NY
Assistant Professor
Department of Mathematics and Statistics *July 2024 - Present*

Smith College Northampton, MA
Postdoctoral Researcher and Lecturer
Department of Mathematical Sciences *July 2022 - June 2024*

Education

Montana State University (MSU) Bozeman, MT
Ph.D. - Mathematics *May 2022*
Thesis: *Directed Graph Descriptors and Distances for Analyzing Multivariate Time Series Data*
Advisor: *Tomáš Gedeon*

Montana State University Bozeman, MT
M.S. - Mathematics *May 2019*

Kenyon College Gambier, OH
B.A. - Mathematics *May 2016*
magna cum laude
Honors and Distinction in Mathematics
Concentration in Scientific Computing

Research Interests

Topological Data Analysis, Computational Topology & Geometry, Mathematics of Data Science.

Publications

Due to working in an interdisciplinary setting, my work follows different conventions for authorship. Mathematics tends to be published alphabetically. Biology tends to be published in descending order of contribution, with students and postdocs listed first, followed by PIs.

11. *Any Graph is a Mapper Graph.* Alvarado, E., **Belton, R.**, Lee, K.J., Palande, S., Percival, S., Tymochko, S., and Purvine, E. To appear in La Matematica. 2026.

10. *Discrete Level Set Persistence for Finite Discrete Functions.* **Belton, R.**, Essl, G. Journal of Foundations of Data Science, Volume 10: 113-157. 2026.

9. *Topology-Preserving Deformations of Streaming Audio via Box Snake Surgery.* Essl, G., **Belton, R.**. In Proceedings of the 50th International Computer Music Conference (ICMC), 2025.

8. *G-Mapper: Learning a Cover in the Mapper Construction.* Alvarado, E., **Belton, R.**, Fischer, E., Lee, K.J., Palande, S., Percival, S., and Purvine, E. SIAM Journal on Mathematics of Data Science, Volume 7(2): 572-596. 2025.

7. *Extremal Event Graphs: A (Stable) Tool for Analyzing Noisy Time Series.* **Belton, R.**, Cummins, B., Fasy, B.T., Gedeon, T., Journal of Foundations of Data Science, Volume 5(1): 81-151. 2023.

6. *Combinatorial Conditions for Directed Collapsing.* **Belton, R.**, Brooks, R., Ebli, S., Fajstrup L., Fasy, B. T., Sanderson, N., & Vidaurre E. Research in Computational Topology 2. Association for Women in Mathematics Series, Volume 30, Springer, Cham. 2022.

5. *Reconstructing Embedded Graphs from Persistence Diagrams.* **Belton, R.**, Fasy, B. T., Mertz, R., Micka, S., Millman, D., Salinas, D., Schenfisch, A., Schupbach, & Williams, L. Computational Geometry Theory and Applications, Volume 90, 2020.

4. *Towards Directed Collapsibility.* **Belton, R.**, Brooks, R., Ebli, S., Fajstrup L., Fasy, B. T., Ray, C., Sanderson, N., & Vidaurre E. Advances in Mathematical Sciences. Association for Women in Mathematics Series, Volume 21. Springer, Cham. 2020.

3. *Learning Simplicial Complexes from Persistence Diagrams.* **Belton, R.**, Fasy, B. T., Mertz, R., Micka, S., Millman, D., Salinas, D., Schenfisch, A., Schupbach, & Williams, L. In Proceedings of Canadian Conference on Computational Geometry (CCCG), 2018.

2. *Loss of TXNIP Enhances Peritoneal Metastasis and Can be Abrogated by Dual TORC1/2 Inhibition.*Spaeth, D., Burch, M., **Belton, R.**, Demoret, B., Grosenbacher, N., David, J., Stets, C., Cohen, D., Shakya, R., Hays, J., & Chen, J. L. Oncotarget, 2018.

1. *A Shape-Context Model for Matching Placental Surface Vascular Networks.* Farnell, E., Farnell, S., Chang, J., Hoffman, M., **Belton, R.**, Keaty, K., Lederman, S., & Salafia, C. Image Analysis & Stereology, 37(1), 55-62. 2018.

Code

1. *Discrete Level Set Persistence for Finite Discrete Functions.* **Belton, R.**, Essl, G. 2025.

2. *G-Mapper: Learning a Cover in the Mapper Construction.* Alvarado, E., **Belton, R.**, Fischer, E., Lee, K.J., Palande, S., Percival, S., and Purvine, E. 2023.

3. *Computing and Comparing Extremal Event DAGs.* **Belton, R.**, Cummins, B., Fasy, B.T., Gedeon, T., Narem, R. 2022.

Book Reviews

1. The Structure and Stability of Persistence Modules, by Chazal, F., Silva, V., Glisse, M., & Oudot, S. Review by **Belton, R.**, and Fasy, B.T., ACM SIGACT News, Vol. 48 Issue 2, June 2017.

Grants, Awards, and Honors

Outstanding Reviewer - 2025. Award to a select number of reviewers of the submissions to the Symposium on Computational Geometry (SoCG).

AMS-Simons Travel Grant - 2023. National award to recent PhD recipients that provided $6000 over two years for research travel expenses.

MSU Student Organization of the Year: Graduate Women in Science and Engineering (WISE) - 2022. Award to one of the 270+ student organizations at MSU that makes significant contributions to the MSU and Bozeman community. I was on the leadership team for WISE from 2020-2022.

William A. Stannard Award for Excellence in Graduate Student Teaching - 2020. $500 award to a graduate teaching assistant in the Department of Mathematical Sciences at MSU.

National Science Foundation Graduate Research Fellowship Program (NSF GRFP) Recipient - 2018. National award for graduate students that provided tuition waivers, covered some research expenses, and provided a $34000 stipend per year for three years.

Outstanding Graduate Student - 2018. Award from the MSU Department of Mathematical Sciences.

STEM Communication Fellowship - 2018. Year long program at MSU that focuses on science communication.

Meritorious Award - 2016. $5000 award to a select number of incoming graduate students at MSU.

Two Pieces of Artwork featured at Bridges Interdisciplinary Art and Mathematics Conference - 2016.

Outstanding Presentation Award - 2015. Award to a select number of undergraduate presenters at Math Fest.

Risk Taker Award - 2015. Award to a math major at Kenyon College.

Presentations

Talks at seminars or colloquia at universities are typically 50 minutes, while my talks at conferences are typically 20-30 minutes.

Invited Talks

45. Topological Data Analysis (TDA) Seminar, Michigan State University. *Studying 2-Parameter Persistent Homology via Directed Topology.* April 2026.
44. Applied Topology in Albany (ATiA) Seminar, University at Albany. *Searching for the Unknown: Parameter Selection in Mapper.* March 2026.
43. Algebra/Topology Seminar, University at Albany. *Studying 2-Parameter Persistent Homology via Directed Topology.* March 2026.
42. Statistical Methodology for Data with Spatial and Temporal Dependencies session at the 19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics). Birkbeck, University of London. *Topological Signal Processing.* December 2025.
41. Innovation Across Disciplines with AI and Data Science, South Dakota State University (Virtual), *Parameter Selection in Topological Data Visualization.* October 2025.
40. Mathematics Colloquium, Union College, *Searching for the Unknown: Parameter Selection in Topological Data Visualization.* October 2025.
39. Symposium for Undergraduate Mathematics Research, SUNY New Paltz, *Searching for the Unknown: Parameter Selection in Topological Data Visualization* (Plenary Talk). September 2025.
38. Joint Statistics Meeting (JSM) in the special session: Novel Methods and Applications for Time Series Data, *Topological Data Analysis for Time Series Data.* August 2025.
37. Joint Meeting of the New Zealand Mathematical Society, Australian Mathematical Society, and AMS in the special session: Applied and Computational Topology, *Discrete Level Set Persistence for Finite Discrete Functions.* December 2024.
36. Applied Algebraic Topology Research Network (AATRN) Online Seminar, *Time Series and Biological Network Analysis via Directed Graphs.* January 2024.
35. Joint Mathematics Meetings (JMM) for the AMS special session: Applied Topology: Theory, Algorithms, and Applications, *Directed Collapsibility and its Connections to Topological Data Analysis.* January 2024.
34. Geometry and Topology Seminar, University of Massachusetts Amherst, *Directed Descriptors of Data.* March 2023.
33. University of Florida Topological Data Analysis Conference, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* February 2023.
32. Joint Mathematics Meetings (JMM) for the AMS special session: Models and Methods for Sparse (Hyper)Network Science, *Towards Automating Covers for Mapper Graphs.* January 2023.
31. Theoretical Biology Seminar, The Pennsylvania State University, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* November 2022.
30. Algebraic and Combinatorial Perspectives in the Mathematical Sciences (ACPMS) Online Seminar, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* November 2022.
29. Sigma Xi Inaugural International Forum for Research Excellence (IFoRE) in the session: The Convergence of Data, Geometry, and Biology: Insights from the 'shape' of Biological Data, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* November 2022.
28. Topology and Data Science Seminar, University of Oklahoma, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* February 2022.
27. Math Colloquium, Kenyon College, *Going Backwards. Reconstructing Shapes from Topological Data Descriptors.* February 2021.
26. Math Colloquium, Augustana University, *Going Backwards. Reconstructing Shapes from Topological Data Descriptors.* September 2020.

25. SIAM Central States Meeting – Minisymposium on Applied and Computational Topology, University of Oklahoma, *Reconstructing Graphs from Persistence Diagrams.* October 2018.

Contributed Talks

24. AMS Eastern Sectional Meeting in the special session on Data Driven Methods in Biology, *Topological Time Series Analysis with Applications in Biology.* April 2025.
23. Young Researchers Forum, Computational Geometry Week, *Any Graph is a Mapper Graph.* June 2024.
22. SIAM Conference on Applied Algebraic Geometry - Minisymposium on Applied Algebraic Topology, Eindhoven University of Technology, *Adaptive Mapper Graphs.* July 2023.
21. Geometry and Topology Meet Data Analysis and Machine Learning Conference, Northeastern University, *Adaptive Mapper Graphs.* June 2023.
20. Hudson River Undergraduate Mathematics Conference, Mount Holyoke College, *Tackling 3 Data Science Problems with a Topological Stone.* April 2023.
19. Topological Data Visualization Workshop, University of Iowa, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* May 2022.
18. Geometry and Topology Meet Data Analysis and Machine Learning Conference, The Ohio State University, *Directed Collapsibility of Euclidean Cubical Complexes.* June 2019.

Local Talks

17. Sigma Xi Tuesday Talks, Smith College, *The Common Misinterpretations and Pitfalls in Topological Data Analysis.* September 2023.
16. Math Colloquium, Smith College, *Tackling 3 Problems with A Topological Stone: My Math Journey.* September 2022.
15. Applied Mathematics Seminar, MSU, *Research Updates.* 2017, 2021, and 2022.
14. Graduate Student Seminar, MSU, *Research Updates.* 2016-2022.
13. Computational Topology and Geometry Seminar, MSU. 2-3 talks per semester on various topics in computational topology and geometry including Reeb spaces, persistent path homology, ε-nets, and algebraic geometry. 2016-2021.
12. Topology and Geometry Seminar, MSU. 1-2 talks per semester on various topics in geometry and topology including lagrangian mechanics, stratified spaces, sheaf theory, and representation theory. 2018-2021.
11. White Dog Brewery, Bozeman, MT, *Everyone has to share. Even the processors in computers.* March 2019.
10. Undergraduate Mathematics Seminar, MSU, *Analyzing Musical Scores Using Topological Data Analysis.* April 2018.
9. Computer Science Seminar, MSU, *Overview of Computational Topology.* September 2017.

Posters

8. Algebraic Topology: Methods, Computation, and Science, Montana State University, *Discrete Level Set Persistence for Finite Discrete Functions.* July 2025.
7. SIAM Conference on Mathematics of Data Science, *Learning Parameters for Mapper Graphs.* October 2024.
6. TDA Week, Kyoto University, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* August 2023.
5. SIAM Conference on Mathematics of Data Science, San Diego, CA, *Extremal Event Graphs: A Stable Tool for Analyzing Noisy Time Series.* September 2022.
4. Algebraic Topology in Data and Dynamics, MSU, *Analyzing Musical Scores Using Topological Data Analysis.* July 2018.
3. Algebraic Topology: Methods, Computation, and Science, IST Austria, *Analyzing Musical Scores Using Topological Data Analysis.* June 2018.
2. Joint Mathematics Meetings, Seattle, WA, *A Shape-Context Model for Matching Placental Surface Vascular Networks.* January 2016.
1. Translational Data Analytics Forum, The Ohio State University, *Identifying Genes of Interest through Clustering and Outlier Analysis of Genomic Time Series Data.* October 2015.

Conference or Workshop Organizer

- DataFest, Vassar College. This event was co-organized with Ming-Wen An, Daryl DeFord, and/or Monika Hu. April 2025, 2026.
- Special Session on Data Driven Methods in Biology, AMS Eastern Sectional Meeting. This event was co-organized with Enrique Alvarado. April 2025.
- Guided Affinity Group (GAG) Leader, SIAM Conference on Mathematics of Data Science. Elizabeth Munch and I mentored a GAG on "Topology and Data". Students in our GAG met with me each morning and attended sessions relevant to our topic. October 2024.
- Organizing Committee of Broader Engagement Program, SIAM Conference on Mathematics of Data Science. I reviewed applications and organized the mentoring program for the conference. October 2024.
- Minisymposium on Topological Data Visualization, SIAM Conference on Mathematics of Data Science. This event was co-organized with Enrique Alvarado and Sarah Percival. October 2024.
- Minisymposium on Applied Algebraic Topology: Theory and Implementation at the SIAM Conference on Applied Algebraic Geometry, Eindhoven University of Technology. This event was co-organized with Elizabeth Munch and Sarah Percival. July 2023.
- WISE Undergraduate Workshop at MSU. A one day workshop for undergraduates interested in graduate school in STEM. The workshop included talks about funding resources, graduate student panels, small group discussions, and mentor-mentee pairings. This event was co-organized with Hannah Goemann and Katrina Lyon. April 2022.
- WISE Conference on COVID-19 Research at MSU. A one day conference free to the public about research and outreach efforts concerning COVID-19. This event was co-organized with Hannah Goemann and Marziah Hashimi. December 2021.

Workshops

Programs I have participated in that either had an active research or skills development component.

Intractability in Discrete Geometry and Topology, Dagstuhl Seminar, 2026.

New Researchers Conference, Vanderbilt University, 2025.

Applied Math in Statistics and Data Science Education, ICERM, 2025.

Preparing to Teach Workshop, Joint Statistical Meetings (JSM), 2024.

Mathematical Challenges in Neuronal Network Dynamics, ICERM, 2023.

Third Workshop for Women in Computational Topology, Swiss Federal Institute of Technology Lausanne (EPFL), 2023.

AMS Mathematical Research Community on Models and Methods for Sparse (Hyper)Network Science, 2022.

Applied Mathematical Modeling with Topological Techniques, ICERM, 2019.

Women in Topology (WIT) Workshop, MSRI, 2017.

Spring School and Conference in Applied and Computational Algebraic Topology, Hausdorff Research Institute for Mathematics (HIM), 2017.

Park City Mathematics Institute (PCMI) Undergraduate Summer School, 2016.

Referee or Reviewer

AWM/Springer volume for the proceedings of the Women in Data Science and Mathematics (WiSDM) workshop

European Symposium on Algorithms (ESA)

European Workshop on Computational Geometry (EuroCG)

Grace Hopper Celebration (GHC) Poster Session

Journal of Applied and Computational Topology (JACT)

Journal of Foundations of Data Science (FoDS)

MathSciNet Mathematical Reviews

SIAM Symposium on Algorithm Engineering and Experiments (ALENEX)

Symposium on Computational Geometry (SoCG)

Symposium on Discrete Algorithms (SODA)

Underrepresented Students in Topology and Algebra Research Symposium (USTARS)

Women in Computational Topology (WinCompTop)

Teaching

Instructor of Record.

Vassar College

M 341: Statistical Inference, Spring: 2025, 2026.

M 301: Topics in Data Science, Fall 2025.

M 244: Intermediate Data Science, Spring: 2025, 2026.

M 241: Probability, Fall 2025.

M 240: Introduction to Statistics, Fall 2024.

Smith College

M 212: Multivariable Calculus, Fall 2023.

M 211: Linear Algebra, Spring 2023.

M 153: Discrete Mathematics, Spring 2024.

M 111: Calculus I, Fall 2022.

Montana State University

M 172: Calculus II, Fall 2019, Spring 2020.

M 171: Calculus I, Fall 2017.

M 121: College Algebra, Fall 2016, Spring 2017.

Teaching Adjacent Experience

Teaching Assistant for Applied Mathematical Modeling with Topological Techniques workshop at ICERM, 2019.

Teaching Assistant for "Learning R" workshops at MSU, 2019.

Tutor at Math Learning Center at MSU for 5 semesters and 1 summer.

Mentoring Experience

- Supervised research focused independent studies at Vassar (3 students total). Fall 2024, 2025, and Spring 2025.
- Research Mentor for *Projects in TDA* course that brings together undergraduate and graduate students from different universities to work on projects in TDA. Project topics include data visualization, dimensionality reduction, and time series analysis. Fall 2024, 2025.
- Met once a week with students (10 total) throughout the academic year and everyday during a 6 week summer period. The project focused on simulating and analyzing plant data using topological and geometric measures. This was co-advised with Chris Golé. 2022-2024.
- Met twice a week with an REU student studying TDA applied to music. The project was supervised by Brittany Terese Fasy and David L. Millman. 2017.
- Regularly assisted math and CS undergraduates prepare presentations for the Computational Topology and Geometry Seminar at MSU.

Service at Vassar

Data Science and Society (DSS) Steering Committee.

Asprey Center for Collaborative Approaches to Science (ACCAS) Steering Committee.

Tenure Track in Statistics Search Committee, Fall 2024.

Visiting Assistant Professor in Statistics Search Committee, Spring 2026.

Leadership Experience

President of Graduate Women in Science and Engineering (WISE) - MSU
WISE aims to support graduate student women and gender minorities in STEM at MSU. In my leadership role, I applied for funding, managed the budget, maintained the website, and organized events such as seminars, article discussions, book clubs, and social events. 2020-2022.

Graduate Program Committee Representative - MSU
Organized social events for the Department of Mathematical Sciences and acted as a liaison between graduate students and faculty. 2018-2020.

President of Computational Topology and Geometry - MSU
Managed the budget and helped organize a weekly seminar for graduate and undergraduate students. 2016-2018.

Invited Panels

Applying to Math Graduate School and REUs, Hudson River Undergraduate Math Conference, 2026.

Applying to Graduate School Panel, Smith College Math Club, 2023.

Picture a Scientist Panel, MSU, 2020.

Graduate student panelist at the Nebraska Conference for Undergraduate Women in Mathematics, 2019.

Outreach

Vassar Science Scholars Program, 2025.

Putnam Problem Co-instructor for Smith College students, 2022.

Volunteer for the Women in Mathematics in New England Conference, Smith College, 2022-2023.

Co-organized Math Craft Workshop at Girls for a Change Workshop at MSU, 2020.

Computational Skills

Certificates. High Performance Computing Certificate from Oak Ridge Computing Course, 2022.

Programming Languages. Python, R, MATLAB, HTML, JavaScript.

Command Line/Version Control. Vim, git.

Software. GitHub, LaTeX, Inkscape, Microsoft.

Operating Systems. Mac, Windows, Linux.

Organizations

American Mathematical Society (AMS)

American Statistical Association (ASA)

Applied Algebraic Topology Research Network (AATRN)

Association for Women in Mathematics (AWM)

Institute of Mathematical Statistics (IMS)

Society for Industrial and Applied Mathematics (SIAM)

Women in Computational Topology (WinCompTop)